

```
#1. Taking Data and creating DataFrame(df) of customers in a mall
import pandas as pd
import numpy as np
df = pd.read_csv("https://raw.githubusercontent.com/ameenmanna8824/DATASETS/main/Mall_Customers.csv")
df
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	
0	1	Male	19	15	39	
1	2	Male	21	15	81	
2	3	Female	20	16	6	
3	4	Female	23	16	77	
4	5	Female	31	17	40	
...	...	...	...	...	...	
195	196	Female	35	120	79	
196	197	Female	45	126	28	
197	198	Male	32	126	74	
198	199	Male	32	137	18	
199	200	Male	30	137	83	

200 rows × 5 columns

```
#2.EDA - Exploratory Data Analysis
df.shape
```

(200, 5)

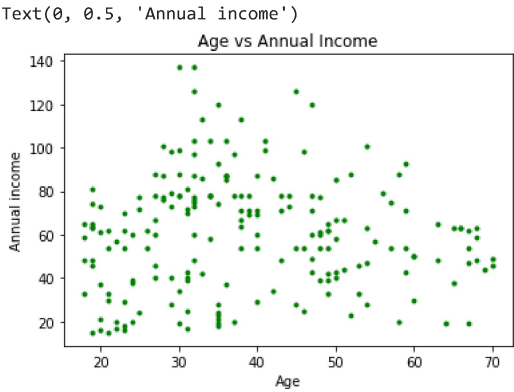
```
df.size
```

1000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   CustomerID            200 non-null   int64  
 1   Genre                 200 non-null   object  
 2   Age                  200 non-null   int64  
 3   Annual Income (k$)    200 non-null   int64  
 4   Spending Score (1-100) 200 non-null   int64  
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
#3.Visualization(Not Required Here)
import matplotlib.pyplot as plt
plt.scatter(df['Age'],df['Annual Income (k$)'], color = "green",marker = ".")
plt.title('Age vs Annual Income')
plt.xlabel('Age')
plt.ylabel('Annual income')
```



```
#4.Divide into Input&Output(x=input,y=output)
#input(x) - Age, Annual Income (in k $)
```

```
#output(y) - Spending Score (1-100)
```

```
x = df.iloc[:,2:4].values
x
```

```
[ 28, 76],
[ 32, 76],
[ 25, 77],
[ 28, 77],
[ 48, 77],
[ 32, 77],
[ 34, 78],
[ 34, 78],
[ 43, 78],
[ 39, 78],
[ 44, 78],
[ 38, 78],
[ 47, 78],
[ 27, 78],
[ 37, 78],
[ 30, 78],
[ 34, 78],
[ 30, 78],
[ 56, 79],
[ 29, 79],
[ 19, 81],
[ 31, 81],
[ 50, 85],
[ 36, 85],
[ 42, 86],
[ 33, 86],
[ 36, 87],
[ 32, 87],
[ 40, 87],
[ 28, 87],
[ 36, 87],
[ 36, 87],
[ 52, 88],
[ 30, 88],
[ 58, 88],
[ 27, 88],
[ 59, 93],
[ 35, 93],
[ 37, 97],
[ 32, 97],
[ 46, 98],
[ 29, 98],
[ 41, 99],
[ 30, 99],
[ 54, 101],
[ 28, 101],
[ 41, 103],
[ 36, 103],
[ 34, 103],
[ 32, 103],
[ 33, 113],
[ 38, 113],
[ 47, 120],
[ 35, 120],
[ 45, 126],
[ 32, 126],
[ 32, 137],
[ 30, 137]]
```

```
y = df.iloc[:,4].values
y
```

```
array([39, 81, 6, 77, 40, 76, 6, 94, 3, 72, 14, 99, 15, 77, 13, 79, 35,
       66, 29, 98, 35, 73, 5, 73, 14, 82, 32, 61, 31, 87, 4, 73, 4, 92,
       14, 81, 17, 73, 26, 75, 35, 92, 36, 61, 28, 65, 55, 47, 42, 42, 52,
       60, 54, 60, 45, 41, 50, 46, 51, 46, 56, 55, 52, 59, 51, 59, 50, 48,
       59, 47, 55, 42, 49, 56, 47, 54, 53, 48, 52, 42, 51, 55, 41, 44, 57,
       46, 58, 55, 60, 46, 55, 41, 49, 40, 42, 52, 47, 50, 42, 49, 41, 48,
       59, 55, 56, 42, 50, 46, 43, 48, 52, 54, 42, 46, 48, 50, 43, 59, 43,
       57, 56, 40, 58, 91, 29, 77, 35, 95, 11, 75, 9, 75, 34, 71, 5, 88,
       7, 73, 10, 72, 5, 93, 40, 87, 12, 97, 36, 74, 22, 90, 17, 88, 20,
       76, 16, 89, 1, 78, 1, 73, 35, 83, 5, 93, 26, 75, 20, 95, 27, 63,
       13, 75, 10, 92, 13, 86, 15, 69, 14, 90, 32, 86, 15, 88, 39, 97, 24,
       68, 17, 85, 23, 69, 8, 91, 16, 79, 28, 74, 18, 83])
```

```
#5.Train & Test Variables
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,random_state = 0) #random_state has been used to avoid data Interchange
```

```
print(x.shape)
print(x_train.shape)
print(x_test.shape)
```

```
(200, 2)
(150, 2)
(50, 2)
```

```
print(y.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(200,)
(150,)
(50,)
```

```
#6.Noramalization/Scaling(only for Inputs)
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
x_train = scaler.fit_transform(x_train)
x_test = scaler.fit_transform(x_test)
```

```
#7.Apply Classifier,Regressor or Cluster
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
```

```
#8.Fit the model
model.fit(x_train,y_train)
```

```
LogisticRegression()
```

```
#9.Predict the Output
y_pred = model.predict(x_test)
y_pred #Predicted Output Values
```

```
array([55, 42, 55, 55, 42, 42, 42, 55, 55, 42, 42, 42, 42, 55, 55, 42, 42,
       42, 42, 55, 42, 42, 42, 42, 55, 42, 42, 42, 42, 55, 42, 42, 42, 42,
       42, 42, 42, 42, 42, 55, 42, 42, 42, 42, 55, 55, 42, 55, 42])
```

```
y_test #Actual Output Values
```

```
array([29, 13, 46, 42, 69, 15, 76, 36, 15, 20, 55, 77, 32, 16, 51, 94, 92,
       9, 73, 47, 88, 97, 65, 73, 56, 91, 90, 97, 58, 28, 35, 41, 17, 54,
       5, 85, 75, 40, 44, 50, 5, 50, 32, 46, 27, 59, 3, 54, 43, 87])
```

```
#10.Accuracy
from sklearn.metrics import accuracy_score
accuracy_score(y_pred,y_test)*100
```

```
0.0
```